

Capstone Project



TrustAI Marketplace Front End/UX
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Introduction

This document turns the MVP into buildable screens and user journeys. It is designed for Sprint 1 UI construction using mock data, then Sprint 2 integration with real backend and AI analysis responses in sprint 3.

This document also defines the comprehensive requirements for developing TrustAI Marketplace.

1 Product objective

The project aims to deliver an AI-assisted decision-support web application that helps users make safer buying decisions on online marketplace listings. The user enters a listing or product description, and the system returns a plain-English analysis that includes a summary, risk indicators, estimated price fairness, suggested seller questions and an overall recommendation.

The application must remain advisory. It must not claim to detect every scam, provide financial guarantees, depend on automated marketplace scraping, or require a custom-trained fraud model for the MVP.

1.2 Requirements-Gathering Discussion

1.2.1 Personas

1.2.1.1 Primary persona: Marketplace buyer

Goal: Assess a marketplace listing before deciding whether to proceed with a purchase.

- A quick and understandable evaluation.
- Clear reasons behind identified risks.
- An indication of whether the asking price appears fair.
- Access to previous listing assessments.

1.2.2 User Stories

1.2.2.1 Epic 1: Auth & Users

1. As a visitor, I can register with email and password so that I can use the service
2. As a user, I can log in and log out so that my session is secure.
3. As a user, I can view and edit my profile.

US-01: Register an account

As a	new buyer
I want	to create an account
So that	I can access TrustAI Marketplace and retain my analyses
Priority	Must have

Acceptance criteria

- Given that I am not signed in, when I open the application, then I can access the registration page.
- Given valid registration details, when I submit the form, then an account is created.
- Given missing or invalid required information, when I submit the form, then the system identifies the affected fields.
- Given an email already associated with an account, when I attempt to register, then the system does not create a duplicate account.
- The user password is never displayed in plain text after submission.
- After successful registration, the user can proceed to the authenticated application.

US-02: Sign in

As a	registered buyer
I want	to sign in securely
So that	I can access my dashboard and saved analyses
Priority	Must have

Acceptance criteria

- Given valid credentials, when I sign in, then I am granted access to the dashboard.
- Given invalid credentials, when I sign in, then access is denied.
- An unauthenticated user cannot access protected application pages.

US-03: Sign out

As a	authenticated buyer
I want	to sign out
So that	Another person cannot continue using my session
Priority	Must have

Acceptance criteria

- When I select sign out, then my authenticated session is terminated.
- After sign-out, protected pages require authentication.
- Using the browser Back button does not restore authenticated access to protected data.

1.2.2.2 Epic 2: Listings & Data

4. As a user, I can submit a listing as pasted text so that it can be analysed.
5. As a user, I can submit a listing URL, and the system fetches the page content.
6. Listings persist in Postgres with lifecycle status (submitted → processing → analysed).
7. Listing parser: extract title, price, description and seller details from text/URL.
8. Validation messages are displayed close to the relevant fields.

US-04: Enter listing information

As a	buyer
I want	to enter marketplace listing details manually
So that	the system can assess the listing
Priority	Must have

Acceptance criteria

- The system identifies which fields are required.
- The price accepts only a valid numeric monetary value.
- Validation messages are displayed close to the relevant fields.
- Invalid submissions are not sent for analysis.

US-05: Provide listing URL

As a	buyer
I want	to include the listing URL
So that	the assessment record can reference the original listing
Priority	Could have

Acceptance criteria

- If a URL is supplied, its format is validated.
- Unsafe URL schemes are rejected.

US-06: Submit a listing for analysis

As a	buyer
I want	to submit completed listing information
So that	I can receive a risk and value assessment
Priority	Must have

Acceptance criteria

- Given valid listing information, when I submit it, then the backend receives the request.
- The submission is associated with the authenticated user.
- If analysis cannot be completed, the system communicates the failure without inventing a result.

1.2.2.3 Epic 3: AI Analysis

9. As a buyer, I see scam indicators with plain-English explanations for my listing.
10. As a buyer, I see a 0–100 risk score combining rule-based and AI signals.
11. As a buyer, I get a recommendation and suggested questions to ask the seller.
12. STRETCH: As a buyer, I can add listing images to the analysis

US-07: Receive a plain-language listing summary

As a	buyer
I want	to receive a concise summary of the listing
So that	I can quickly understand its main characteristics
Priority	Must have

Acceptance criteria

- A completed assessment includes a summary.
- The summary is based on the submitted information.
- The summary is displayed separately from risk indicators and recommendations.

US-08: View a risk score and reasons

As a	buyer
I want	to see the listing risk level and supporting reasons
So that	I understand why caution may be necessary
Priority	Must have

Acceptance criteria

- A completed assessment presents a risk score or clearly defined risk level.
- The result includes one or more reasons supporting risk assessment.
- Reasons are displayed in understandable language.

US-09: View estimated price fairness

As a	buyer
I want	to receive an indication of price fairness
So that	I can recognise a potentially unrealistic asking price
Priority	Must have

Acceptance criteria

- The completed result includes a price-fairness assessment of when sufficient input is available.
- The result provides a short reason or basis for the fairness indication.
- If information is insufficient, the system indicates that fairness could not be determined.
- The system does not present the estimate as financial advice or a guarantee.

US-10: Receive suggested seller questions

As a	buyer
I want	to receive suggested questions for the seller
So that	I can obtain more information before deciding
Priority	Must have

Acceptance criteria

- The result includes suggested seller questions.
- Questions are relevant to the submitted listing and identified risks.
- Questions are shown as suggestions rather than evidence that the seller is dishonest.
- The user can read and copy the questions.
- Missing seller questions causes the response to be treated as incomplete unless the team intentionally makes the field optional.

US-11: Receive an overall recommendation

As a	buyer
I want	to receive a Buy, Caution, or Avoid recommendation
So that	I can quickly understand the overall assessment
Priority	Must have

Acceptance criteria

- A completed result contains exactly one supported recommendation value: Buy, Caution, or Avoid.
- The recommendation is visually distinguishable without relying only on colour.
- The application clearly states that the result is advisory.

1.2.2.4 Epic 4: Saved history

US-14: View saved analysis history

As a	returning buyer
I want	to view previous analyses
So that	I can revisit earlier assessments
Priority	Must have

Acceptance criteria

- The user can open a history page from the authenticated application.
- History contains only records the user is authorized to access.
- Selecting a record displays its listing information and analysis.
- A clear empty state appears when the user has no saved analyses.

1.3 User Journey

1.3.1 New buyer analyses a listing.

Stage	User action	System response	User value	Exception
1. Discover	Open TrustAI Marketplace	Display landing page and sign-in or registration options	Understand the advisory purpose	Application unavailable
2. Register	Enter account details	Validate details and create account	Gain access to protected features	Duplicate email, weak password, invalid details
3. Sign in	Submit credentials	Authenticate and open dashboard	Reach a private workspace	Invalid credentials, locked account, service error
4. Start analysis	Select Analyze listing	Display listing form	Begin an assessment	Form fails to load
5. Enter listing	Provide title, price, currency, source, description and optional URL	Validate required fields and formats	Know whether submission is complete	Missing title, invalid price, unsupported input
6. Submit	Submit listing	Create pending analysis and send request	Receive progress feedback	Duplicate submission, timeout, service unavailable
7. Analyze	Waiting for processing	Produce validated structured output and pricing/rules indicators	Receive a consistent assessment	Invalid or incomplete AI response
8. Review result	Read summary, risks, price assessment, questions and recommendation	Explain results in plain language	Make a more informed decision	Some components unavailable
9. View history	Open history and select result	Display authorized previous analyses	Revisiting past decisions	No records, deleted records, access denied
10. Sign out	Select sign out	Terminate session	Protect account information	Session termination fails

1.3.2 Returning buyer reviews analysis history.

- The buyer signs in.
- The buyer opens saved analysis history.
- The system retrieves only analyses belonging to that buyer.
- The system shows a list of previous submissions.
- The buyer selects an analysis.

1.3.3 Core End-to-End User Flow

Open TrustAI → Sign in / Register → Submit listing details → Validate form
→ Analyse listing → Show loading state → Display results → View history
→ User can copy seller questions, analyse another listing or view history

1.4 Information Architecture and Navigation

Information Architecture and Navigation

Unauthenticated user

- └ Login
- └ Register
- └ Forgot password

Authenticated user

- └ Submit Listing / Dashboard
- └ Results
- └ History
- └ Account / Logout

Navigation rule

All pages except Login, Register and Forgot Password should be protected. After successful login, the user should land on Submit Listing because submission is the core MVP action.

1.5 Requirements

1.5.1 Functional Requirements

The requirements Have a traceability matrix linking requirements to original user stories and journey phases for verification

1.5.1.1 Account and access management (FUNC-AAM)

ID	Proposed requirement
FUNC-AAM-01	Allow a new user to register an account.
FUNC-AAM-02	Allow a registered user to sign in and sign out.
FUNC-AAM-03	Associate listings and analyses with the submitting user.

1.5.1.2 Listing management (FUNC-LM)

ID	Proposed requirement
FUNC-LM-04	Provide fields for title, price, currency, source, and description.
FUNC-LM-05	Validate required fields and permitted formats before analysis.
FUNC-LM-06	Transmit a valid listing to the backend analysis endpoint.
FUNC-LM-07	Provide visible feedback while analysis is processing.

1.5.1.3 Analysis (FUNC-A)

ID	Proposed requirement
FUNC-A-08	Produce a plain-language listing summary.
FUNC-A-09	Produce a risk score or level and supporting reasons.
FUNC-A-10	Provide one overall recommendation: Buy, Caution, or Avoid.
FUNC-A-11	Validate the AI service structured response before displaying or saving it as complete.
FUNC-A-12	Display a clear advisory limitation statement.

1.5.1.4 History (FUNC-H)

ID	Proposed requirement
FUNC-H-13	Allow users to view their previous analyses.

1.5.2 Non-Functional Requirements

1.5.2.1 Performance (NF-PERF)

ID	Proposed requirement
NF-PERF-01	Standard pages should become usable within 4 seconds under normal demonstration conditions, excluding external AI-processing time.
NF-PERF-02	The API should acknowledge a valid analysis submission within 2 seconds, then process it according to the approved design.
NF-PERF-03	Queries for an ordinary user history view should complete within 2 seconds under the agreed MVP test load.
NF-PERF-04	Define and test an expected concurrent-user demonstration load before final acceptance.

1.5.2.2 Usability and Reliability (NF-USR)

ID	Proposed requirement
NF-USR-05	Use plain language for summaries, risk explanations, errors, and advisory notices.
NF-USR -06	Support desktop and common mobile-width browsers through a responsive layout.
NF-USR -07	Provide visible labels, clear validation messages, and logical keyboard order.
NF-USR -08	Do not communicate risk level or recommendation by color alone.
NF-USR -09	Make processing, success, empty and failure states distinguishable and understandable.

1.5.2.3 Maintainability (NF-MAIN)

ID	Proposed requirement
NF-MAIN-10	Define clear interfaces and responsibilities for frontend, backend, AI, data, and deployment concerns.
NF-MAIN-11	Externalize configuration from application code.
NF-MAIN -12	Include setup instructions, service structure, tests, and deployment information in the repository.
NF-MAIN -13	Automate tests for core business logic, validation, authorization, and API contracts.
NF-MAIN -14	Require accepted changes to pass local and CI checks.
NF-MAIN -15	Update architecture, API, setup, and test documentation when affected.

1.5.2.4 Security (NF-SEC)

ID	Security requirement
NF-SEC-01	Never log passwords, authentication tokens, API keys, or secrets.
NF-SEC -02	Do not disclose whether an email address is registered in authentication errors.
NF-SEC -03	Expire authenticated sessions after an approved inactivity period and invalidate them at sign-out.
NF-SEC -04	Protect authentication endpoints against automated guessing through rate limiting or equivalent controls.

1.6 Traceability Summary

Screen-level mapping from wireframe to underlying epic and user stories, for quick cross-reference during review.

Screen	Epic	User Stories	Requirement families
Auth	Epic 1: Auth & Users	US-01, US-02, US-03	FUNC-AAM-01, FUNC-AAM-02, FUNC-AAM-01, NF-SEC -03
Submit Listing	Epic 2: Listings	US-04, US-05, US-06	FUNC-LM-04 - 05
Results	Epic 3: AI Analysis	US-07 to US-11	FUNC-A-08-012, NF-PERF-01, NF-USR-05
History	Epic 4: Saved History	US-14	FUNC-H-13, NF-PERF-03

1.7 Trust AI Wireframes with Requirement Annotations

Fig 1. Auth/Login Page

TRUSTAI MARKETPLACE - WIREFRAMES
01 / Auth - low-fi, Sprint 1 handoff

TrustAI Marketplace
Safer buying decisions for online marketplace users

Why use TrustAI

- AI listing summary
- Risk indicators
- Seller questions
- Saved history

Login form

[SIGN IN](#) [REGISTER](#)

Email *

name@example.com

Password *

[SIGN IN](#)

[Forgot password?](#) [New user? Create account](#)

Security copy -
TrustAI stores analysis history for signed-in users.

States: empty (value prop + form) - validation error (field-level msg) - invalid credentials (non-technical msg, stay on page) - success (redirect to Submit Listing)

Login form with email/password fields, inline error messaging, and redirect-on-success to Submit Listing.

- FUNC-AAM-01: Allow a new user to register an account.
- FUNC-AAM-02: Allow a registered user to sign in and sign out.

Fig 2. Submit Page

TRUSTAI MARKETPLACE — WIREFRAMES
02 / Submit Listing — low-fi, Sprint 1 handoff

trustai.app/submit

TrustAI Marketplace / Submit [SUBMIT](#) [HISTORY](#) [ACCOUNT](#) [LOGOUT](#)

Submit a marketplace listing for analysis

Listing details

Title *
iPhone 14 Pro 256GB

Price * €500 **Currency *** ZAR R

Source *
Facebook Marketplace
e.g. Facebook Marketplace, Gumtree, OLX

Optional URL
https://_

Description *
Paste listing text

Tips for better analysis

- Include full description
- Include price and seller claims
- Add URL if available
- Do not include private data

ANALYZE LISTING

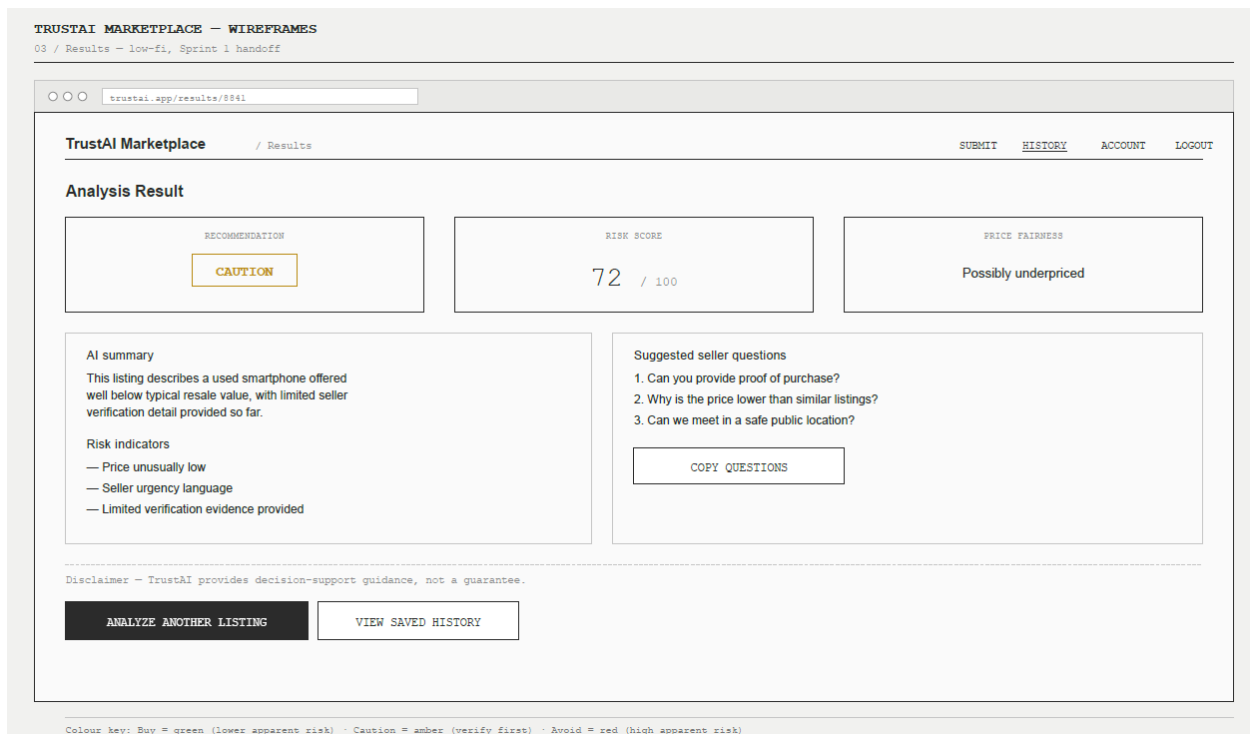
State: Ready — CTA enabled once required fields are valid

States: ready · loading ("Analysing listing...", duplicate submit disabled) · error (friendly msg, input preserved) · success (routes to Results)

Form for listing title, description, price, and source URL, with inline validation and an Analyze button that triggers a loading state.

- FUNC-LM-04: Provide fields for title, price, currency, source, and description.
- FUNC-LM-05: Validate required fields and permitted formats before analysis.
- FUNC-LM-06: Transmit a valid listing to the backend analysis endpoint.
- FUNC-LM-07: Provide visible feedback while analysis is processing.

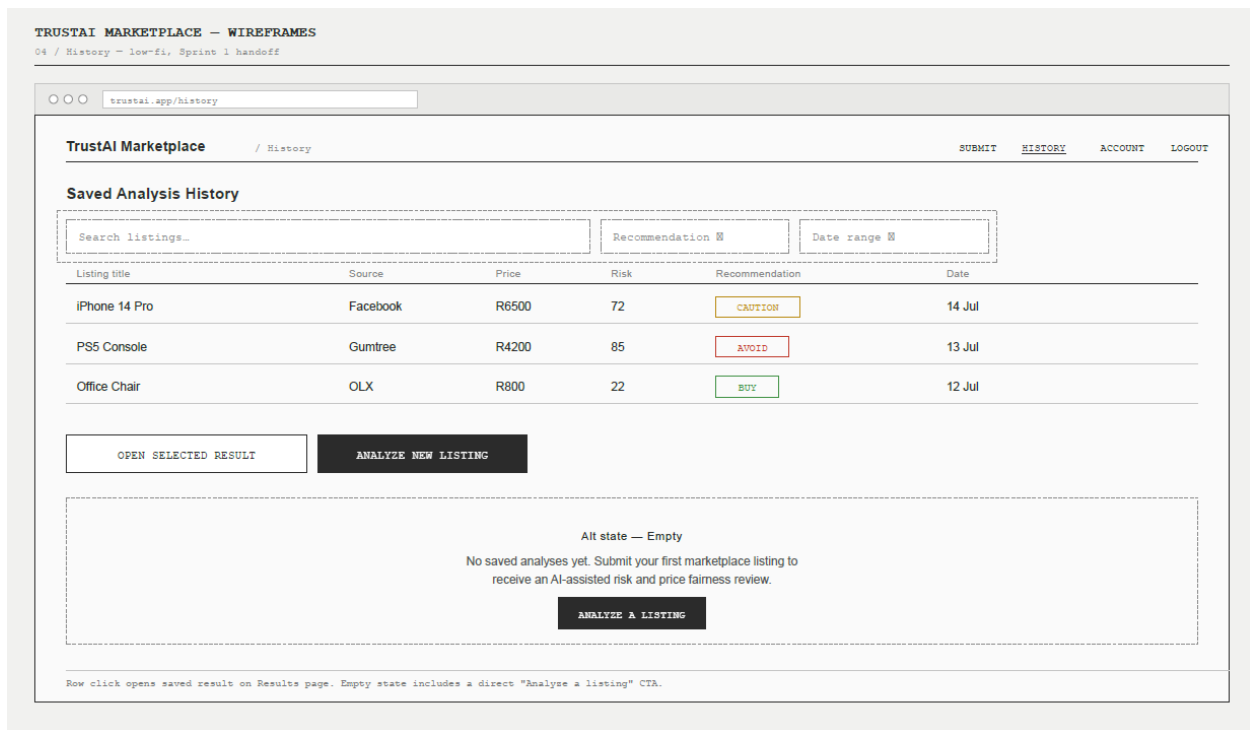
Fig 3. Results Page



RiskGauge component visualizing risk_score and analysis.risk_level, with listing_source and a plain-language explanation displayed alongside.

- FUNC-A-08: Produce a plain-language listing summary.
- FUNC-A-09: Produce a risk score or level and supporting reasons.
- FUNC-A-10: Provide one overall recommendation: Buy, Caution, or Avoid.
- FUNC-A-12: Display a clear advisory limitation statement.
- NF-PERF-012: Results render within 1s of response.

Fig 4. History Page



List of past submissions (date, listing title, risk level) with the ability to reopen any entry's full Results view.

- FUNC-H-13: Allow users to view their previous analyses.

1.8 Shared Component Inventory

Component	Used on	Notes
Top navigation bar	All protected pages	Show TrustAI brand, Submit, History, Account and Logout
Protected route wrapper	Submit, Results, History	Redirect unauthenticated users to Login
Button	All pages	Primary and secondary variants
Form input	Auth, Submit	Support validation states
Recommendation badge	Results, History	Colour-coded risk communication
Risk score card	Results, History	Numeric score with meaning
History table row	History	Clickable row opens saved result
Disclaimer banner	Submit, Results	Avoid guarantees and set user expectations

1.9 Frontend/API Contract

Endpoint	Method	Frontend trigger	Expected output
/auth/register	POST	Register form submit	Created user/session or validation error
/auth/login	POST	Login form submit	Session/token and user object
/analysis	POST	Analyze Listing button	Structured analysis response
/analysis/history	GET	History page load	Saved analyses for current user
/analysis/{id}	GET	Open saved result	Full saved analysis details

Submit Analysis Request JSON

```
{
  "title": "iPhone 14 Pro 256GB",
  "price": 6500,
  "currency": "ZAR",
  "source": "Facebook Marketplace",
  "url": "https://example.com/listing",
  "description": "Seller states phone is sealed, urgent sale, cash only..."
}
```

Analysis Response JSON

```
{
  "summary": "The listing describes a high-value smartphone at a discounted price.",
  "riskScore": 72,
  "recommendation": "Caution",
  "riskIndicators": [
    "Price appears unusually low",
    "Seller uses urgent sale language",
    "Limited verification evidence provided"
  ],
  "priceAssessment": "Possibly underpriced compared with typical market value.",
  "sellerQuestions": [
    "Can you provide proof of purchase?",
    "Can we meet in a safe public location?",
    "Can you share the device serial number or IMEI for verification?"
  ]
}
```

2.0 Acceptance Criteria

Area	Acceptance criteria
Auth	Register, login, invalid login error, redirect after success and protected-page enforcement
Submit	Required fields, optional URL, validation, loading state and structured backend request
Results	Recommendation, risk score, summary, reasons, price note, seller questions and saved result
History	Saved analyses list, open saved result, empty state and optional search/filter support

Conclusion

By defining clear requirements, including product catalog, shopping cart and checkout, payment integration, order tracking, security, scalability, and performance — this document establishes a foundation for development, testing, and deployment.

The successful implementation of these requirements will result in a reliable, user-friendly, and enhance customer satisfaction, and provides a framework for future enhancements.